

推导完全分类

我会先界定旧回测到底否证了哪个简化子系统，再逐项裁决完整检验必须接入哪些语义层：区间套、类型分层、正规出场、账本与选择偏差。

According to a document from 2026-07-01, the current dispute is exactly that the backtest chain did not exercise several mechanisms that the formal system treats as essential: `schedule_adapter` never emits `EnterEarning`, so the `EarningShares` phase is structurally unreachable, and the TW layer has only raw transition legality but no trigger predicate for when recovery or earning should occur.

The strict ruling is:

The old “no alpha” results are valid only for the simplified implemented strategy, not for full

They are not useless; they refute specific simplified objects. But they cannot be lifted to:

“complete Chanlun has no alpha.”

That lift would require a theorem that the simplified strategy and full strategy produce the same trade set, same state labels, same exits, same capital schedule, and same PnL functional. Your own audit says they do not.

0. Formal setup

Let a backtest strategy be the tuple

$$\Pi = (\Gamma, Z, A, E, M, \mathcal{K}, J),$$

where:

Γ_t = signal / certificate generator,

Z_t = state-classification map,

A_t = action interpreter,

E_t = exit rule,

M_t = capital / sizing / treasury rule,

$\mathcal{K}_t$ = risk-feasible set,

J_t = projection objective.

The realized incremental PnL is a functional

$$R(\Pi; P) = \sum_t [N_t^\Pi(P_{t+1} - P_t) - C_t^\Pi],$$

or, for per-signal tests,

$$X_i^\Pi = \delta_i(P_{\tau_i^{out}} - P_{\tau_i^{in}}) - C_i.$$

An alpha statement is:

$$\mathbb{E}[R(\Pi; P)] > 0$$

or, conditionally,

$$\mu_{\Pi}(z, a) = \mathbb{E}[X^{\Pi}(a) \mid Z^{\Pi} = z] > 0.$$

The old tests evaluated some simplified strategy

$$\Pi_{\text{simp}}.$$

The full Chanlun strategy is

$$\Pi_{\text{full}}.$$

To infer from old tests that full Chanlun has no alpha, one would need:

$$R(\Pi_{\text{simp}}; P) = R(\Pi_{\text{full}}; P) \quad \text{for all price paths } P.$$

Equivalently, at least:

$$\Gamma_{\text{simp}} = \Gamma_{\text{full}}, \quad Z_{\text{simp}} = Z_{\text{full}}, \quad A_{\text{simp}} = A_{\text{full}}, \quad E_{\text{simp}} = E_{\text{full}}, \quad M_{\text{simp}} = M_{\text{full}}.$$

Your audit says several equalities fail. Therefore:

$$R(\Pi_{\text{simp}}) \leq 0 \text{ \textcolor{red}{centernot}} \Rightarrow R(\Pi_{\text{full}}) \leq 0.$$

This is the core proof.

Q1. Are the previous alpha rejections invalid?

Ruling

$$\text{They are invalid as rejections of full Chanlun.}$$

$$\text{They remain valid as rejections or diagnostics of the simplified object actually tested.}$$

Confidence: **High**.

Proof

Old tests refute:

$$H_{\text{simp}} : \mathbb{E}[R(\Pi_{\text{simp}})] > 0.$$

But the intended target is:

$$H_{\text{full}} : \mathbb{E}[R(\Pi_{\text{full}})] > 0.$$

A rejection of H_{simp} implies rejection of H_{full} only if:

$$\Pi_{\text{simp}} = \Pi_{\text{full}}$$

in the relevant PnL sense.

But your five missing mechanisms change the strategy components:

$$P1 : \Gamma_{\text{simp}} \neq \Gamma_{\text{full}},$$

because the full strategy should use recursive interval-nesting certificates

$$N_{\ell \downarrow e}^{\delta},$$

while the simplified implementation uses only the base-case/single-level signal.

$$P2 : \Gamma_{\text{simp}} \neq \Gamma_{\text{full}}$$

or at least the candidate domain differs, because MACD one-vote veto removes non-MACD candidates before they can enter the empirical sample.

$$P4 : Z_{\text{simp}} \neq Z_{\text{full}},$$

because collapsing $B_1, B_2, B_3, S_1, S_2, S_3$ into only δ destroys type-conditioned returns.

$$P7 : E_{\text{simp}} \neq E_{\text{full}},$$

because "exit at next reverse signal" is not equal to Chanlun's typed exits such as first-class sell close or third-class sell reduce.

$$GAP3 : M_{\text{simp}} \neq M_{\text{full}},$$

because the three-stage treasury process never reaches `EarningShares`; the document explicitly states that `schedule_adapter` emits only `ShortDiff` / `RecoverCapital`, never `EnterEarning`.

Therefore:

$$\Pi_{\text{simp}} \neq \Pi_{\text{full}}.$$

Hence:

$$\mathbb{E}[R(\Pi_{\text{simp}})] \leq 0 \not\Rightarrow \mathbb{E}[R(\Pi_{\text{full}})] \leq 0.$$

That proves the old negative results cannot be promoted to "full Chanlun has no alpha."

What still survives?

Some diagnostics remain useful:

1. The finding that odd/even alternation can be a holding-window identity remains valid for that exit口径, not as a full-system no-alpha proof. Your own report says the alternation was driven by confirmation and "hold until next reverse signal," not by independent structure prediction.
2. The finding that single-instrument BTC beta contaminates raw PnL remains valid. The alpha-extraction report says BTC net results were largely beta rather than independent structural alpha.
3. Any result that is purely about the simplified exit rule, simplified signal universe, or simplified bucket key remains valid only inside that simplified object.

When would Q1 flip?

It would flip only if you prove an equivalence theorem:

$$\boxed{\Pi_{\text{simp}} \equiv \Pi_{\text{full}}}$$

in the sense that they generate identical entries, exits, sizes, costs, treasury states, and PnL on every path.

Given P1/P4/P7/GAP3, that theorem is false unless those mechanisms are shown to be alpha-irrelevant by a separate invariant, for example:

$$\Gamma_{\text{full}} \setminus \Gamma_{\text{simp}} \text{ has zero feasible trades or zero conditional value.}$$

No such invariant is currently given.

Q2. Is $P1 + P4 + P7$ the minimal sufficient set for a valid old-Chanlun single-instrument alpha test?

Ruling

$P1 + P4 + P7$ is a strong minimal core for the buy/sell-point alpha line.

But:

$P2$ is required if the claim includes “all legitimate divergence candidates.”

And:

$GAP3$ is required if the claim includes full Chanlun treasury / negative-cost share accumula

Confidence: **Medium-high.**

Formal reason

For a per-signal buy/sell alpha test, the estimand is:

$$\mu(z, a) = \mathbb{E}[X(a) \mid Z = z].$$

The three minimal components for that estimand are:

P1: interval-nesting certificate

Without recursive interval nesting,

$$N_{\ell \downarrow e}^\delta$$

is replaced by its degenerate base layer:

$$Conf_e^\delta.$$

Then the signal universe is:

$$\Gamma_{\text{simp}} = Conf_e^\delta$$

instead of:

$$\Gamma_{\text{full}} = \{\gamma : N_{\ell \downarrow e}^\delta(\gamma) = 1\}.$$

Since:

$$N_{\ell \downarrow e}^\delta = Cand_\ell^\delta \wedge [J_{\ell-1}^\delta \subseteq J_\ell^\delta] \wedge N_{\ell-1 \downarrow e}^\delta,$$

the full signal set generally differs from the single-level set. Therefore $P1$ is necessary.

P4: type-preserving buckets

If the state key is only:

$$z = (\ell, \delta),$$

but the full system needs:

$$z = (\ell, \delta, I_\gamma, \dots), \quad I_\gamma \subseteq \{1, 2, 3\},$$

then:

$$\mu(\ell, \delta) = \sum_I \mu(\ell, \delta, I) \mathbb{P}(I \mid \ell, \delta).$$

A positive B_1 and a negative B_2 can cancel in the pooled bucket. Thus $P4$ is necessary for typed alpha.

P7: canonical exit

The return functional is exit-dependent:

$$X_i^E = \delta_i(P_{\tau_i^E} - P_{t_i}) - C_i.$$

If the simplified exit is E_4 and canonical Chanlun exit is E_C , then generally:

$$X_i^{E_4} \neq X_i^{E_C}.$$

Therefore a rejection under E_4 does not reject E_C . Thus $P7$ is necessary.

So $P1 + P4 + P7$ is a minimal core for:

single-instrument, buy/sell-point, no treasury alpha.

Where P2 fits

P2 changes the candidate universe:

$$\Gamma_{\text{MACD}} \subseteq \Gamma_{\text{force}} \subseteq \Gamma_{\text{structural}}.$$

If MACD is a hard veto, the test estimates:

$$\mathbb{E}[X \mid \gamma \in \Gamma_{\text{MACD}}],$$

not:

$$\mathbb{E}[X \mid \gamma \in \Gamma_{\text{structural}}].$$

Thus P2 is not optional if the claim is:

full Chanlun divergence candidates have / have no alpha.

It is optional only if the narrowed claim is:

MACD-defined Chanlun candidates have / have no alpha.

Where GAP3 fits

GAP3 is not needed for a pure per-signal BSP alpha test. But it is necessary for the full strategy including "negative cost earns shares," because that feature is specifically implemented through `EarningShares`, which the report says is unreachable in the current schedule adapter.

Thus:

$$P1 + P4 + P7$$

is minimal only for the old-style signal alpha test.

For complete Chanlun strategy alpha:

$$P1 + P2 + P4 + P7 + GAP3$$

is the minimal sufficient set.

When would Q2 flip?

It would flip if:

1. interval nesting is proven observationally equivalent to base-case signals;
2. buy/sell types are proven conditionally exchangeable:

$$\mu(\ell, \delta, B_1) = \mu(\ell, \delta, B_2) = \mu(\ell, \delta, B_3);$$

3. canonical exits are proven PnL-equivalent to reverse-signal exits;
4. MACD is declared as the canonical definition of divergence;
5. treasury dynamics are explicitly outside the target claim.

Q3. Which account model: A, B, or C?

Ruling

Choose C: dual-layer accounting.

R account and TW account should coexist with a formally specified interface.

Confidence: **High**.

Why

You have two different semantic objects:

R account

$$R_t = \Pi_t - A_t - W_t.$$

It is a closed-loop trading-performance account: profits, costs, withdrawals.

TW account

$$TW_t = free_t + holding_t + withdrawn_t.$$

It is a treasury/wealth-stage account. The uploaded TW report states that Lean already proves preservation of TW, stage monotonicity, and OQ-9 invariants, while Rust has bit-exact parity.

These are not substitutes.

R answers:

How much realized trading surplus remains after cost and withdrawals?

TW answers:

What phase of capital recovery / share accumulation is the system in?

A single ledger cannot easily preserve both meanings without either:

1. losing physical withdrawal semantics W_t , or
2. contaminating trading PnL with treasury states.

Thus define a projection/interface:

$$\varphi : R_t \rightarrow TW_t$$

or an event bridge:

$$event_t \in \{ShortDiff, RecoverCapital, Withdraw, EnterEarning, BuyCore\}.$$

Then require interface invariants:

$$Withdraw(w) : \quad W_{t+1} = W_t + w, \quad withdrawn_{t+1} = withdrawn_t + w.$$

$$RecoverProfit(g) : \quad \eta_{t+1} = \eta_t + g.$$

$$BuyCore(a) : \quad \eta_{t+1} = \eta_t + g - a.$$

The three-stage report explicitly identifies that `schedule_adapter` currently does not derive the `EnterEarning` event, which means this bridge is incomplete.

Formal proof that C dominates A/B

Let:

$$\mathcal{S}_R$$

be the state needed for trading PnL, and:

$$\mathcal{S}_{TW}$$

the state needed for treasury phase invariants.

A-only loses some TW predicates, e.g.:

$$TStage = III, \quad openLegacyLegs = 0, \quad withdrawn \geq I_0.$$

B-only loses or distorts trading PnL decomposition:

$$\Pi, A, W, R.$$

C uses:

$$\mathcal{S}_C = \mathcal{S}_R \times \mathcal{S}_{TW}$$

with consistency constraints. Therefore it contains both projections:

$$\pi_R(\mathcal{S}_C) = \mathcal{S}_R,$$

$$\pi_{TW}(\mathcal{S}_C) = \mathcal{S}_{TW}.$$

So any invariant expressible in A or B is expressible in C, while not conversely. Thus C is strictly more expressive and avoids conflating semantics.

When would Q3 flip?

A could be chosen if TW is downgraded to mere observation and not part of strategy behavior.

B could be chosen if R is redefined entirely inside TW and all trading PnL quantities are recoverable as TW projections.

Neither is currently supported by the reported implementation state.

Q4. Should η_* be state-dependent or constant?

Ruling

$$\eta_* \text{ should be state-dependent: } \eta_*(x_t) = L_{t+1}^{wc}(x_t) + \kappa Q_t.$$

A constant is acceptable only as a conservative global upper bound:

$$\eta_* \geq \sup_x \{L_{t+1}^{wc}(x) + \kappa Q_t(x)\}.$$

Confidence: **High** for formal systems; **Medium** for engineering simplicity.

Derivation

The PDF budget constraint in your uploaded report is:

$$\begin{aligned} \eta_{n+1} &= \eta_n + g_n - a_n, \\ a_n + L_{n+1}^{wc} &\leq \eta_n + g_n - \eta_*. \end{aligned}$$

If requiring per-share negative cost at least κ , the report states:

$$\begin{aligned} \eta_t &\geq \kappa Q_t, \\ a_n + L_{n+1}^{wc} + \kappa \Delta Q_n &\leq \eta_n + g_n - \kappa Q_n. \end{aligned}$$

The role of η_* is not mystical; it is a safety barrier. At time t , to allow a new purchase a_t , the post-trade buffer must cover worst-case loss and desired negative-cost cushion:

$$\eta_{t+1} = \eta_t + g_t - a_t \geq L_{t+1}^{wc} + \kappa(Q_t + \Delta Q_t).$$

Thus pre-trade available spend must satisfy:

$$a_t + L_{t+1}^{wc} + \kappa \Delta Q_t \leq \eta_t + g_t - \kappa Q_t.$$

So the natural state-dependent barrier is:

$$\eta_*(x_t) = L_{t+1}^{wc}(x_t) + \kappa Q_t.$$

If a constant is used, it must dominate the state-dependent barrier over the allowed state space:

$$\eta_*^{const} \geq \sup_{x \in \mathcal{X}^{allow}} [L_{t+1}^{wc}(x) + \kappa Q_t(x)].$$

Otherwise the constant can be too low in high-risk/high-position states or too high in low-risk states.

How to choose κ

κ is a risk-policy parameter, not derivable from price alone.

It can be set by one of these principles:

1. Per-share negative-cost floor

$$c_t^{adj} \leq -\kappa.$$

Since:

$$c_t^{adj} = -\eta_t/Q_t,$$

this gives:

$$\eta_t \geq \kappa Q_t.$$

2. Stress drawdown tolerance

Choose κ such that under stress move s_t :

$$\kappa Q_t \geq q_t s_t.$$

3. Margin/risk budget

Set κ so that post-trade stress loss plus margin reserve remains below allowed surplus.

Thus:

κ is canonical only relative to an explicit risk principle.

When would Q4 flip?

Use a constant if your Lean/Rust invariants require finite-state simplicity and you can prove it dominates the state-dependent barrier. Otherwise state-dependent is the stricter canonical formulation.

Q5. Can interval nesting be implemented gradually?

Ruling

Engineering can be gradual; semantic acceptance cannot be downgraded.

Single-level `DivCand` can be a milestone, but it cannot honestly be called "interval nesting integrated."

Confidence: **High**.

Formal definition

The full certificate is:

$$N_{e \downarrow e}^\delta = Conf_e^\delta,$$

$$N_{\ell \downarrow e}^\delta = Cand_\ell^\delta \wedge [J_{\ell-1}^\delta \subseteq J_\ell^\delta] \wedge N_{\ell-1 \downarrow e}^\delta.$$

Thus:

$$N_{\ell \downarrow e}^\delta = 1$$

implies the whole chain:

$$J_e^\delta \subseteq J_{e+1}^\delta \subseteq \cdots \subseteq J_\ell^\delta.$$

A single-level gate is:

$$G_\ell^\delta = \text{Cand}_\ell^\delta.$$

Clearly:

$$N_{\ell \downarrow e}^\delta \Rightarrow G_\ell^\delta.$$

But generally:

$$G_\ell^\delta \not\Rightarrow N_{\ell \downarrow e}^\delta.$$

Counterexample:

$$\text{Cand}_\ell^\delta = 1$$

but either:

$$\text{Cand}_{\ell-1}^\delta = 0$$

or:

$$J_{\ell-1}^\delta \not\subseteq J_\ell^\delta.$$

Then:

$$G_\ell^\delta = 1$$

but:

$$N_{\ell \downarrow e}^\delta = 0.$$

Therefore single-level `DivCand` is a necessary but not sufficient condition for the full interval-nesting certificate.

Correct staged implementation

Use milestone names honestly:

1. `DivCandGate` : only Cand_ℓ^δ .
2. `TwoLevelNest` : $\text{Cand}_\ell^\delta \wedge J_{\ell-1} \subseteq J_\ell \wedge \text{Conf}_{\ell-1}$.
3. `FullNest` : full $N_{\ell \downarrow e}^\delta$.

Only stage 3 may be accepted as:

区间套已接入.

When would Q5 flip?

Only if you explicitly redefine the target strategy as a single-level divergence gate, not Chanlun interval nesting. That is a different hypothesis.

Q6. Is MACD one-vote veto a real selection bias?

Ruling

Yes. It is a sample-selection mechanism embedded into the label definition.

Confidence: **High**.

Proof

Let the structural candidate universe be:

$$\mathcal{U}.$$

Let the MACD-veto candidate set be:

$$\mathcal{U}_M = \{\gamma \in \mathcal{U} : MACD(\gamma) = 1\}.$$

The test estimates:

$$\mu_M = \mathbb{E}[X \mid \gamma \in \mathcal{U}_M].$$

But the desired full structural estimand may be:

$$\mu_U = \mathbb{E}[X \mid \gamma \in \mathcal{U}].$$

Since:

$$\mathcal{U}_M \subseteq \mathcal{U},$$

generally:

$$\mu_M \neq \mu_U.$$

The missing set:

$$\mathcal{U} \setminus \mathcal{U}_M$$

is never observed in the trade sample. Therefore you cannot evaluate:

$$\mathbb{E}[X \mid \gamma \in \mathcal{U} \setminus \mathcal{U}_M].$$

This is selection bias.

It is not necessarily look-ahead bias, but it is a **definition-induced sample restriction**. If MACD is not the canonical Chanlun definition of divergence, the test only evaluates "MACD-Chanlun," not Chanlun divergence.

Correct fix

Three acceptable fixes, in decreasing order of statistical cleanliness:

Fix 1: Broad candidate universe + features

Define *Cand* structurally:

$$Cand(\gamma) = 1$$

if direction, comparability, new high/low, and structural divergence context hold.

Then include MACD / Force / volume / power as features:

$$Z = (\dots, MACDDiv, ForceDiv, PowerRatio, \dots).$$

The selector learns:

$$\chi = 1[LCB(\mu(Z, a)) > \theta].$$

This avoids pre-deleting candidates.

Fix 2: Multiple Θ versions

Run:

$$\Theta_{MACD}, \quad \Theta_{Force}, \quad \Theta_{Hybrid}$$

as distinct pre-registered strategies and compare OOS.

Fix 3: MACD canonical claim

If you insist MACD is the official definition, then the estimand is only:

$$\mu_{MACD}.$$

You must state:

We test MACD-defined Chanlun, not full force-divergence Chanlun.

When would Q6 flip?

Only if MACD area is declared and justified as the canonical definition of `DivCand`. Your own technical definition says M_Θ may be MACD or force accumulation, and flags MACD one-vote veto as concerning; so under the current project goal, Q6 is real.

Q7. Is “install P1/P2/P4/P7/GAP3 and rerun” the correct path to test full Chanlun alpha?

Ruling

Yes, with one refinement: you must predefine the estimand and split signal-alpha from treas

Confidence: **High**.

Correct full-path architecture

There are at least three alpha layers:

Layer 1: Signal alpha

$$\mu(z, a) = \mathbb{E}[X(a) \mid z].$$

Requires:

$$P1 + P2 + P4 + P7.$$

Layer 2: Portfolio alpha

$$\mathbb{E} \left[\sum_t N_t (P_{t+1} - P_t) - C_t \right] > 0.$$

Requires signal layer plus sizing/risk projection.

Layer 3: Treasury / negative-cost share accumulation value

Requires GAP3 and TW/R bridge.

Because `EnterEarning` is currently unreachable, this third layer has not been tested.

The three-stage document says the issue is precisely that `schedule_adapter` never emits `EnterEarning`; it also says the PDF gives explicit stage boundaries involving η_* and $W_t \geq I_0$.

Thus rerunning after installation is the right path, but you must report three separate outcomes.

Formal acceptance tests

Signal-level test

For each typed interval-nested certificate:

$$z = (\ell, \delta, I_\gamma, \text{nest depth}, \text{Cand type}, \text{exit type}, \dots),$$

estimate:

$$\mu(z, a).$$

Use OOS, block bootstrap, and LCB:

$$LCB_{95}(\mu(z, a)) > 0.$$

Strategy-level test

Run full portfolio policy and compare to baseline:

$$\Delta R = \sum_t [\Delta N_t(P_{t+1} - P_t) - \Delta C_t].$$

Use market-neutral / permutation / multi-asset checks if claiming structural alpha beyond beta. Existing reports already warn that raw BTC PnL is strongly beta-contaminated and that odd/even alternation can be a holding-window identity.

Treasury-stage test

Check path properties:

$$\mathbb{P}(\exists t : S_t = III) > 0,$$

$$Q_T > Q_0,$$

$$W_T \geq I_0,$$

$$\eta_T \geq \eta_\star,$$

and portfolio value versus risk/cost baseline.

Do not mix treasury effects with raw signal μ .

Methodological hazards after “full install”

Even after installing everything, false positives and false negatives remain possible.

False positives

1. Too many Θ versions without OOS correction.
2. Choosing MACD/Force/hybrid after seeing results.
3. Picking only positive buy/sell types after in-sample sorting.
4. BTC beta contamination.
5. Reusing the same data to tune $\eta_\star, \kappa, \text{exit}$.

False negatives

1. Full interval nesting too restrictive due to settled-only candidates.

- 2. MACD hard veto deleting valid force-divergence candidates.
- 3. Exit rule too late.
- 4. High-level signals too sparse.
- 5. Treasury value showing up as path utility, not simple μ .

So the rerun must be pre-registered:

Θ fixed before OOS.

Final rulings summary

Question	Ruling	Confidence	Flip condition
Q1 old alpha rejections	Invalid for full Chanlun; valid only for simplified tested object	High	Prove simplified and full PnL-equivalent
Q2 minimal sufficient set	P1+P4+P7 for signal alpha core; add P2 for full divergence; add GAP3 for full strategy/treasury	Med-high	Narrow target excludes divergence breadth or treasury
Q3 ledger	Choose C dual-layer	High	Collapse TW into R or R into TW with proven projection
Q4 η_*	State-dependent barrier $L^{wc} + \kappa Q$; constant only as global conservative bound	High	Simplicity prioritized and constant dominance proven
Q5 interval nesting	Gradual implementation OK; semantic acceptance requires full $N_{\ell \downarrow e}^\delta$	High	Redefine target as single-level gate
Q6 MACD veto	Real selection bias unless MACD is declared canonical	High	MACD accepted as canonical definition
Q7 full rerun	Correct path, but split signal alpha, portfolio alpha, treasury value	High	If target is only simplified MACD/single-level/reverse-exit system

The most compact final statement is:

Old tests refuted Π_{simp} , not Π_{full} .

$\Pi_{\text{full}} = \Pi(P1, P2, P4, P7, GAP3)$

has a different certificate set, state space, return functional, exit map, and treasury transition system. Therefore its alpha must be tested anew.



